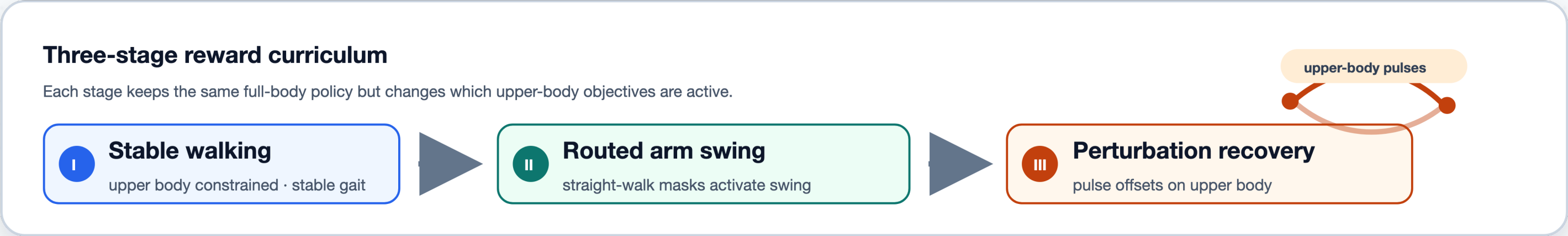
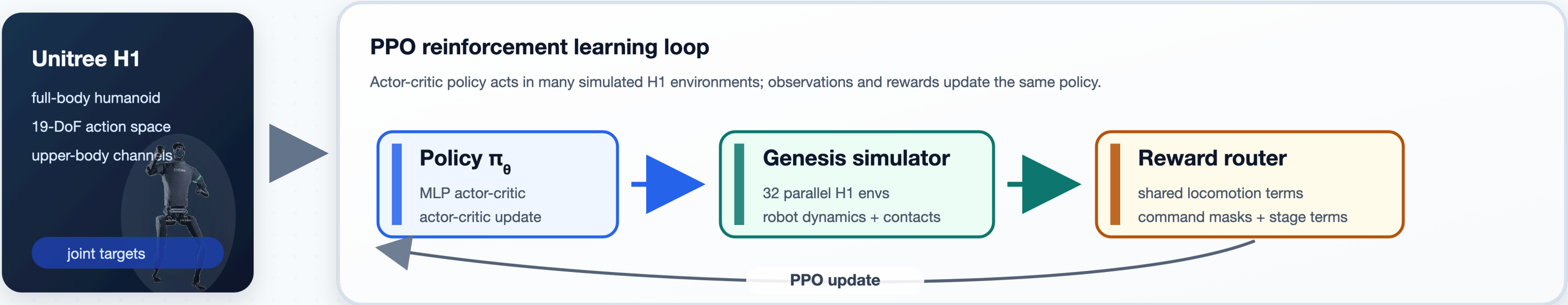


METHOD OVERVIEW

Staged PPO Curriculum for Full-Body H1 Arm-Swing Locomotion

A single 19-DoF policy is trained in Genesis, then shaped through staged reward routing and upper-body recovery.



$$\mathbf{r}_t = \mathbf{r}_{\text{track}} + \mathbf{r}_{\text{safe}} + m_{\text{stand}} \mathbf{r}_{\text{stand}} + m_{\text{walk}} \mathbf{r}_{\text{walk}} + m_{\text{straight}} \mathbf{r}_{\text{swing}} + \mathbf{r}_{\text{robust}}$$

● shared terms preserve locomotion

● straight-walk routing enables visible arm swing

● Stage III trains perturbation recovery